

# Project-3 :Design, Deployment and Monitoring of a Highly Available Web Application on AWS

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**Project:**Design Deployment and Monitoring of a Highly Available Web Application on AWS.

## Objective

- Login AWS
- Project Purpose

## Architecture Structure

- Structure
- Explanation

## AWS Services Used

- Amazon EC2
- Elastic Load Balancing (Application Load Balancer)
- Auto Scaling Group
- Amazon EBS
- Amazon VPC
- AWS IAM
- Amazon CloudWatch
- Amazon SNS

## Testing Outcome

- Uploaded
- Created

## Conclusion

- Summary
- Learning

## 1. OBJECTIVE-

- Login to AWS Console
- Search EC2
- Open to EC2 Dashboard

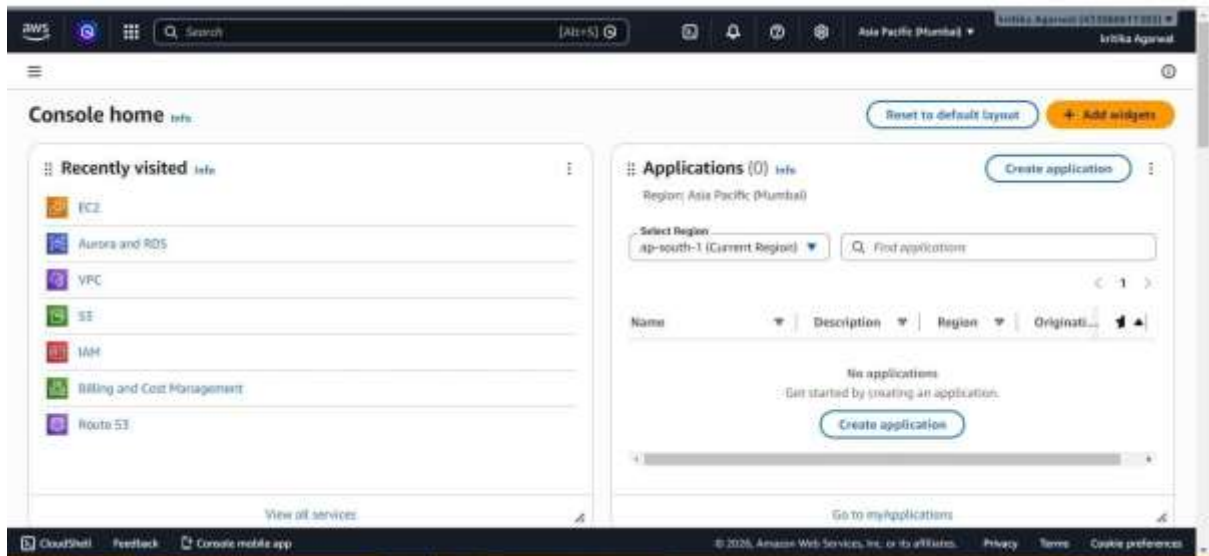


FIG: AWS CONSOLE

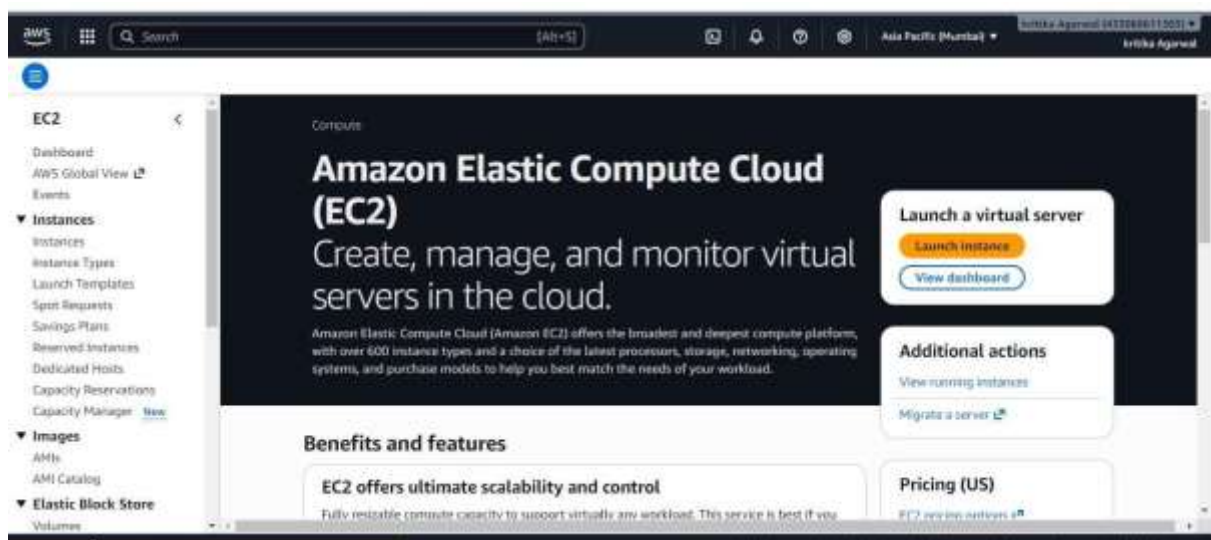


FIG: EC2 BOARD

## 2. ARCHITECTURE STRUCTURE-

- User sends a request through the Internet.
- ALB receives the request.
- ALB forwards traffic to healthy EC2 instances in AZ-1 or AZ-2.
- Auto Scaling adjusts the number of EC2 instances based on traffic.
- CloudWatch monitors the infrastructure.
- SNS sends alerts if CPU usage or any configured metric crosses the threshold.

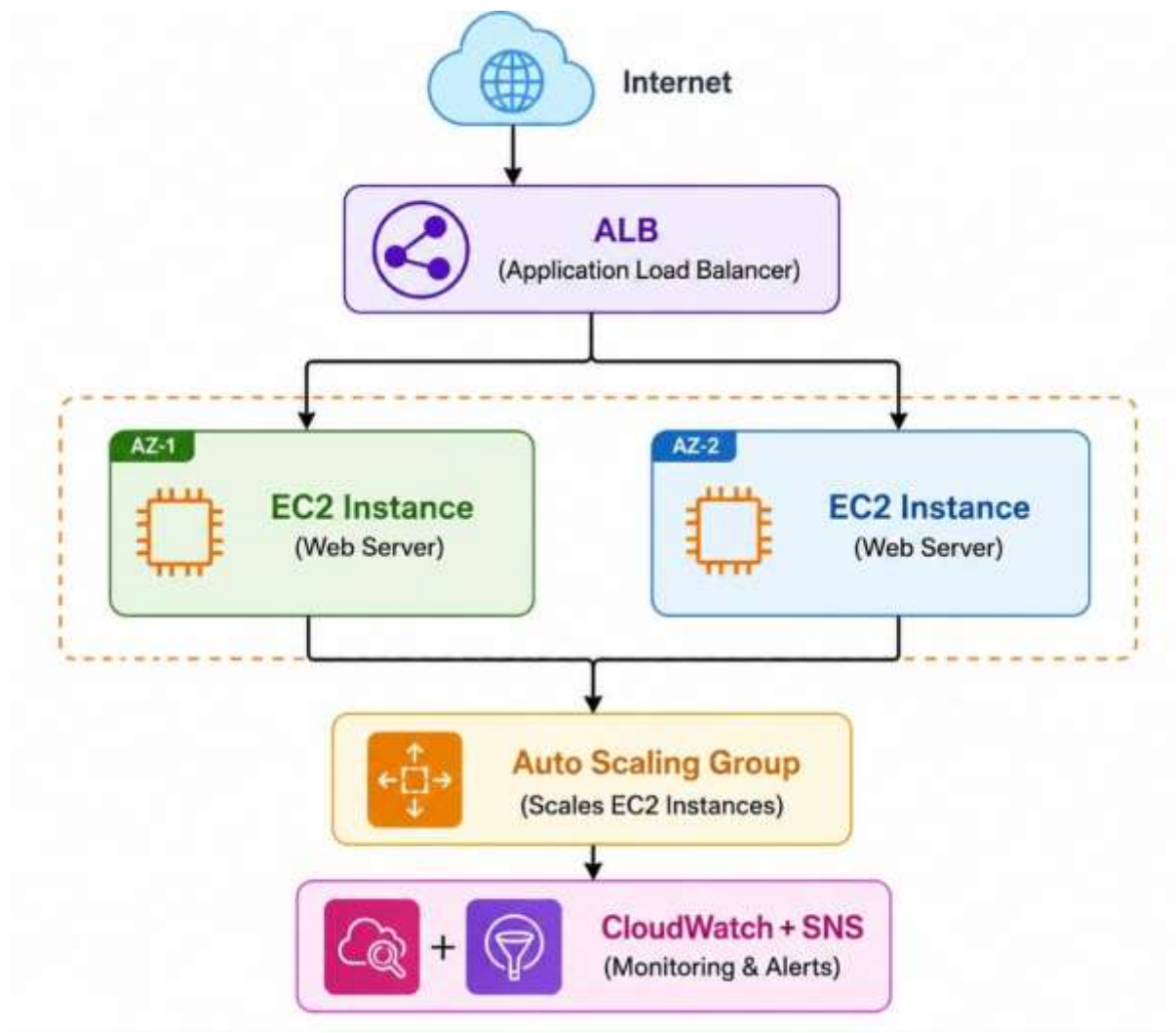


FIG: WEB ARCHITECTURE

### 3. AWS SERVICES USED-

#### ❖ AWS Setup and Security:

#### Step 1: Create an AWS Account:

- Sign in to the AWS Management Console.
- Select the AWS Region.
- Understand Availability Zones.

#### Step 2: Configure IAM

- Open the IAM Console.
- Create a new IAM User.
- Assign only the required permissions.
- Enable Multi-Factor Authentication (MFA).

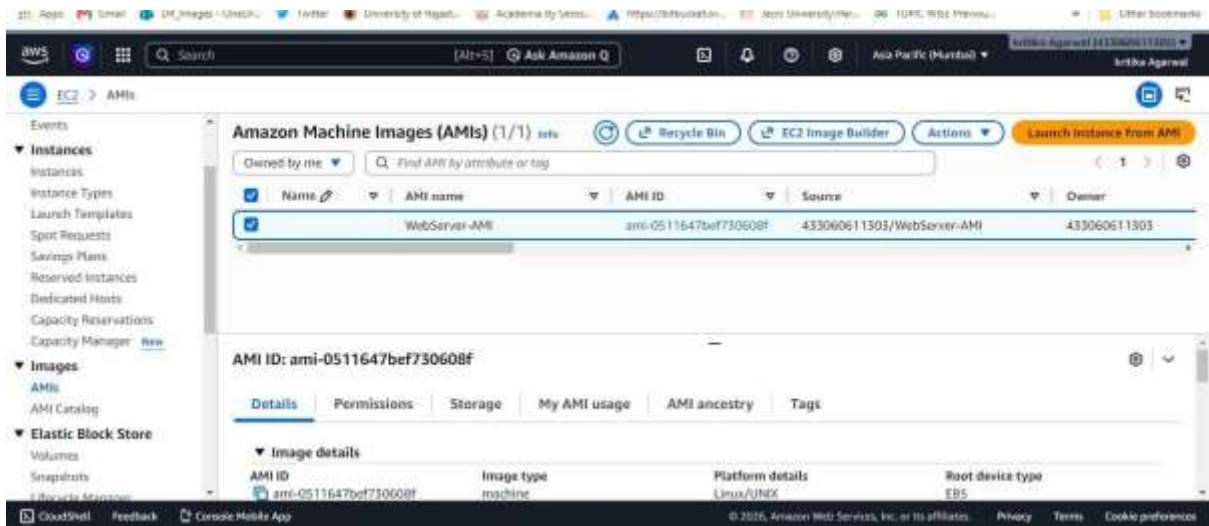
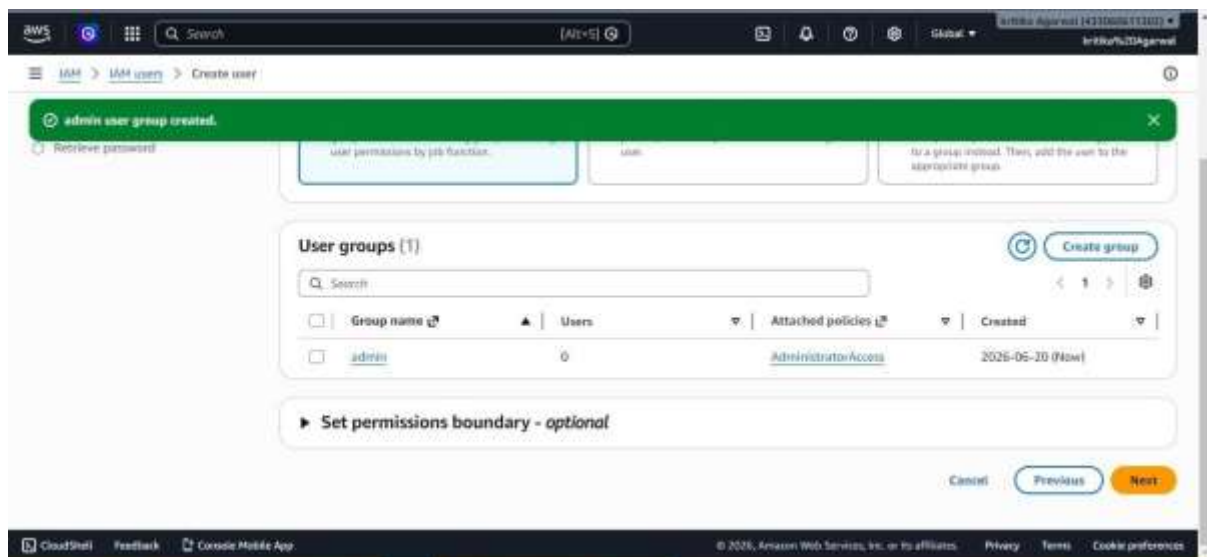


FIG: AMI

FIG: IAM USER



## ❖ Network Design:

### Step 3: Create a VPC

- Open the VPC Dashboard.
- Create a new VPC.
- Create Public Subnets in multiple Availability Zones.
- Attach an Internet Gateway.
- Configure Route Tables.
- Configure Security Groups for HTTP and SSH access.

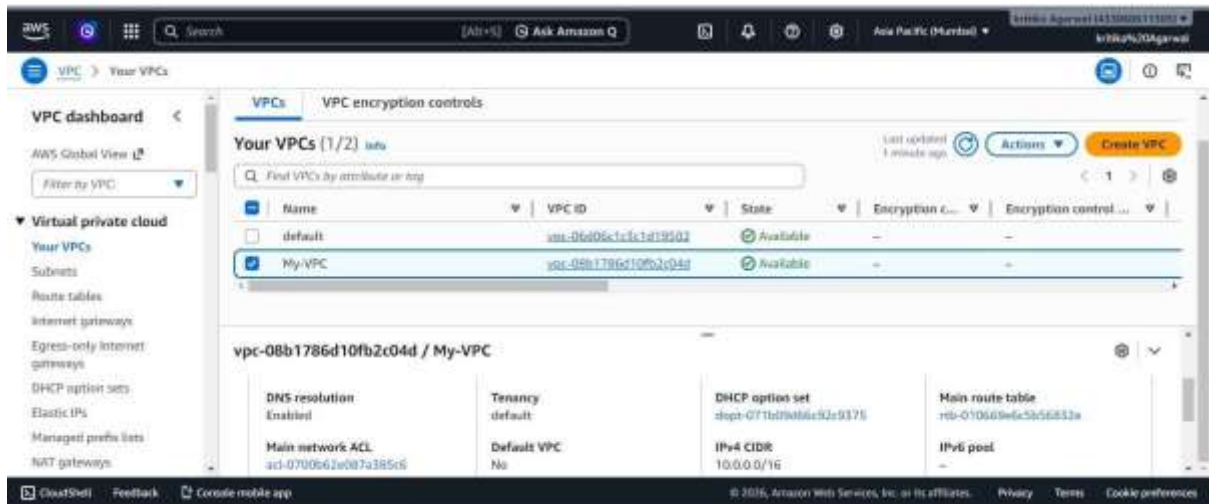


FIG:VPC

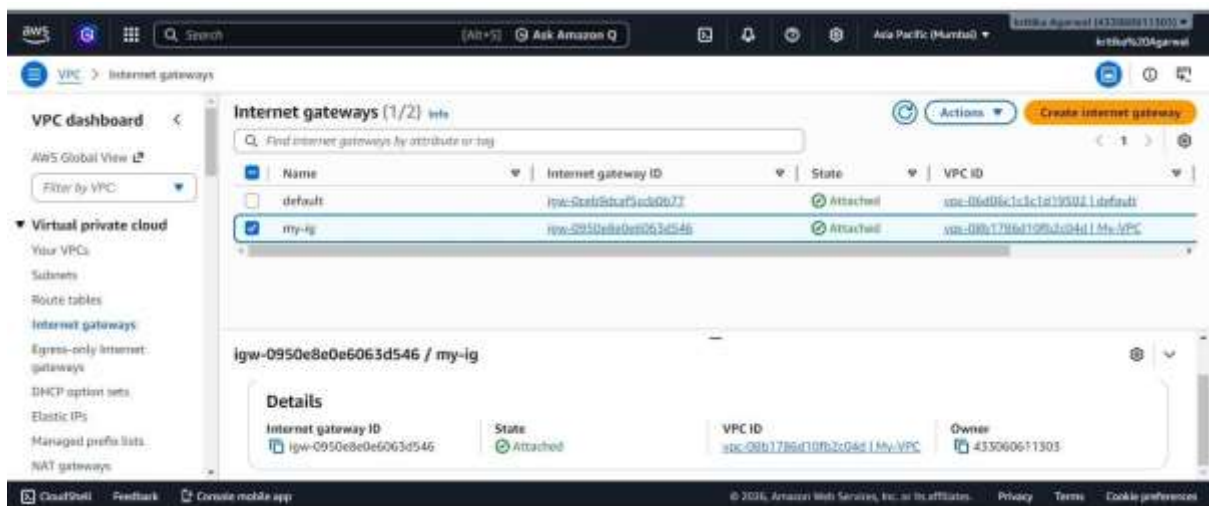


FIG: IGW

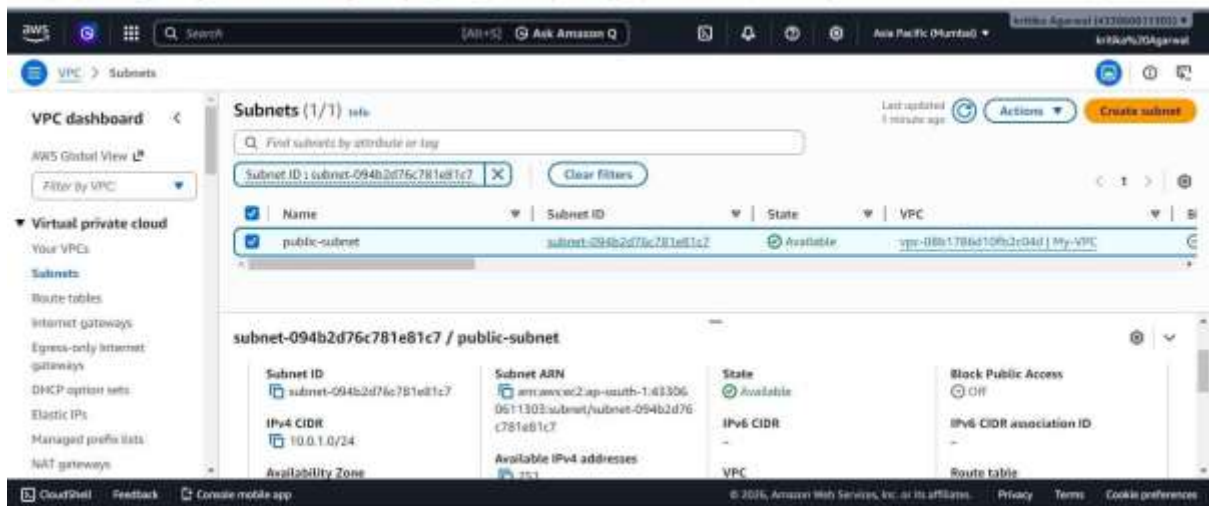


FIG: PUBLIC SUBNET

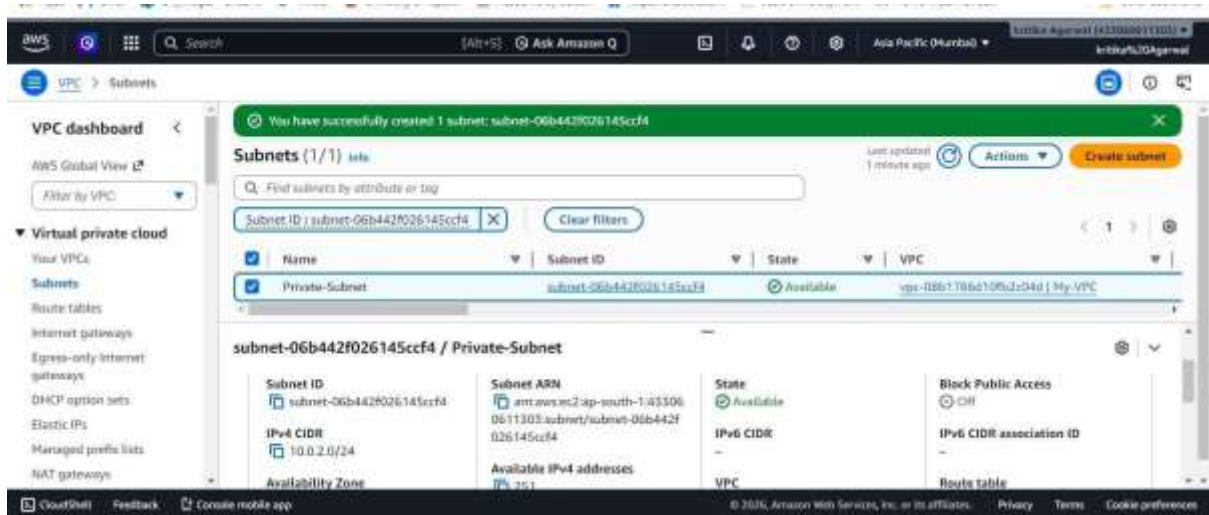


FIG: PRIVATE SUBNET

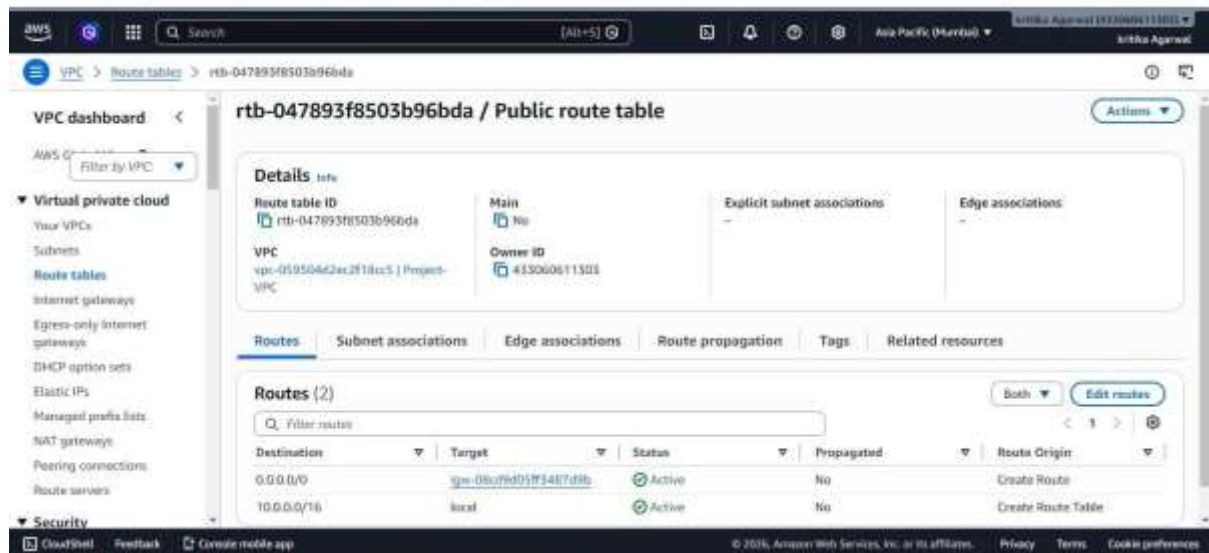


FIG: PUBLIC RT

### ❖ Compute and Storage:

#### Step 4: Launch an EC2 Instance

- Launch an Amazon Linux EC2 instance.
- Select the instance type.
- Configure the Security Group.
- Connect using SSH.
- Install Apache Web Serve.
- ❖ Run commands:
  - `sudo yum update -y`
  - `sudo yum install httpd -y`
  - `sudo systemctl start httpd`
  - `sudo systemctl enable httpd`

FIG: EC2 INSTANCE

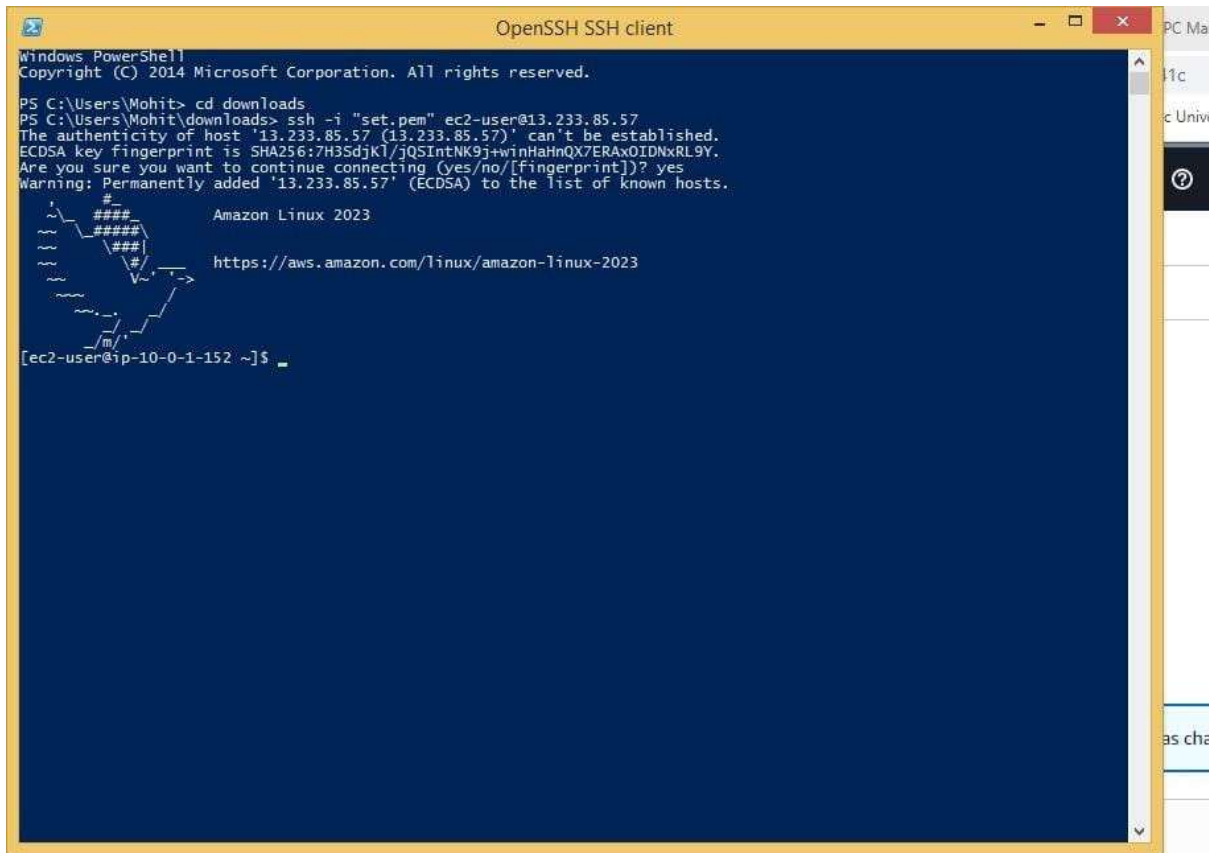
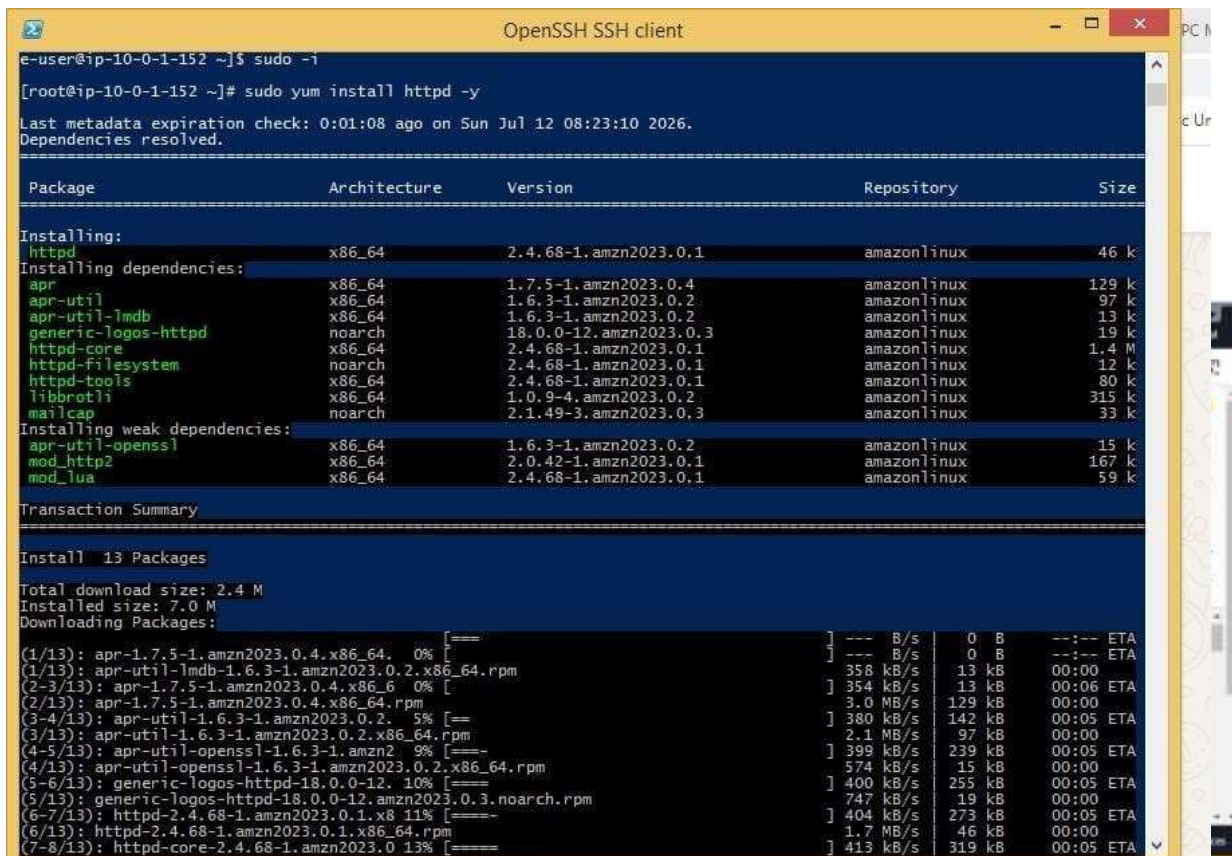


FIG: APACHE INSTALL





- Create a new EBS Volume.
- Attach the volume to the EC2 instance.
- Format and mount the volume.
- Verify that data remains available after restarting the instance.



## ❖ High Availability and Scalability:

### Step 6: Create an Application Load Balancer:

- Create a Target Group.
- Register EC2 instances.
- Create an Application Load Balancer.
- Attach the Target Group.
- Verify that traffic is distributed among EC2 instances.

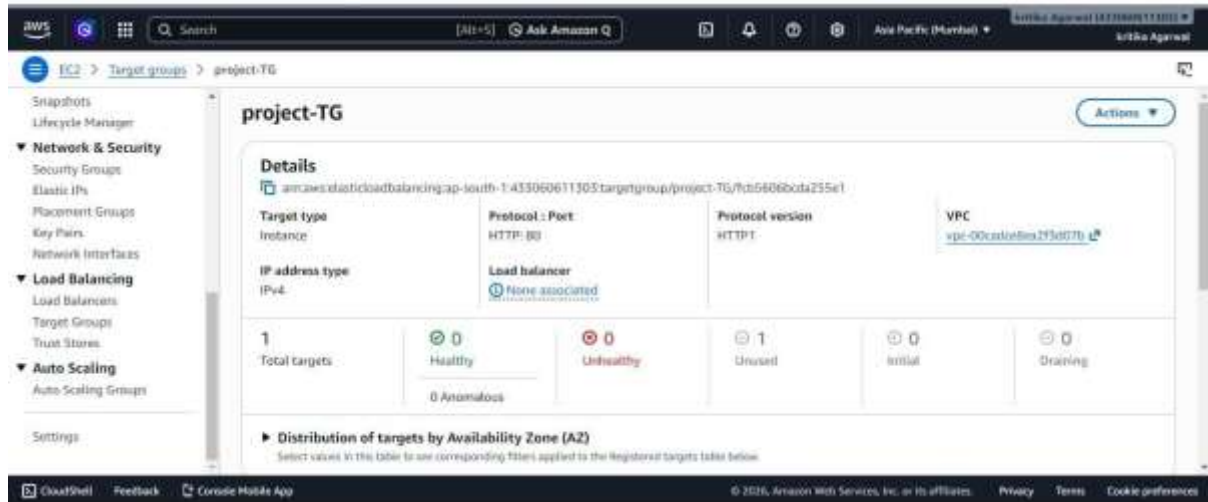


FIG: TG

### Step 7: Configure Auto Scaling Group:

- Create a Launch Template.
  - Create an Auto Scaling Group.
  - Configure:
    1. Minimum Capacity
    2. Desired Capacity
    3. Maximum Capacity
- Create a Target Tracking Scaling Policy.
  - Set CPU Utilization as the scaling metric.

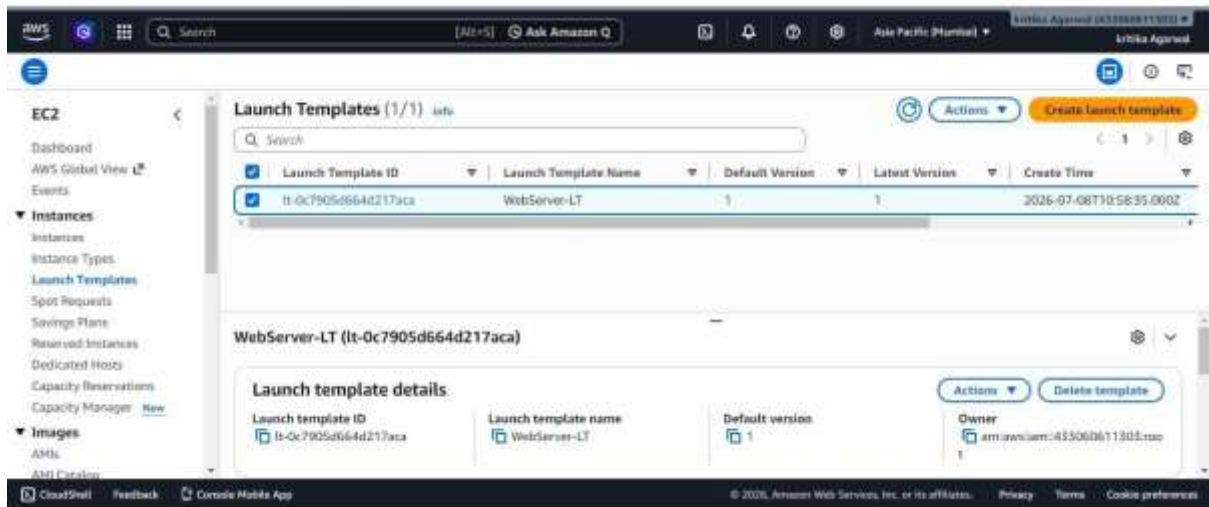


FIG: LAUNCH TEMPLATE

### ❖ Monitoring and Observability:

## Step 8: Configure Amazon CloudWatch

1. Open the CloudWatch Console.

2. Create a CloudWatch Alarm.

3. Monitor:

- CPU Utilization
- EC2 Status Check
- Network Metrics

4. Set the alarm threshold.

CloudWatch continuously monitors the application.

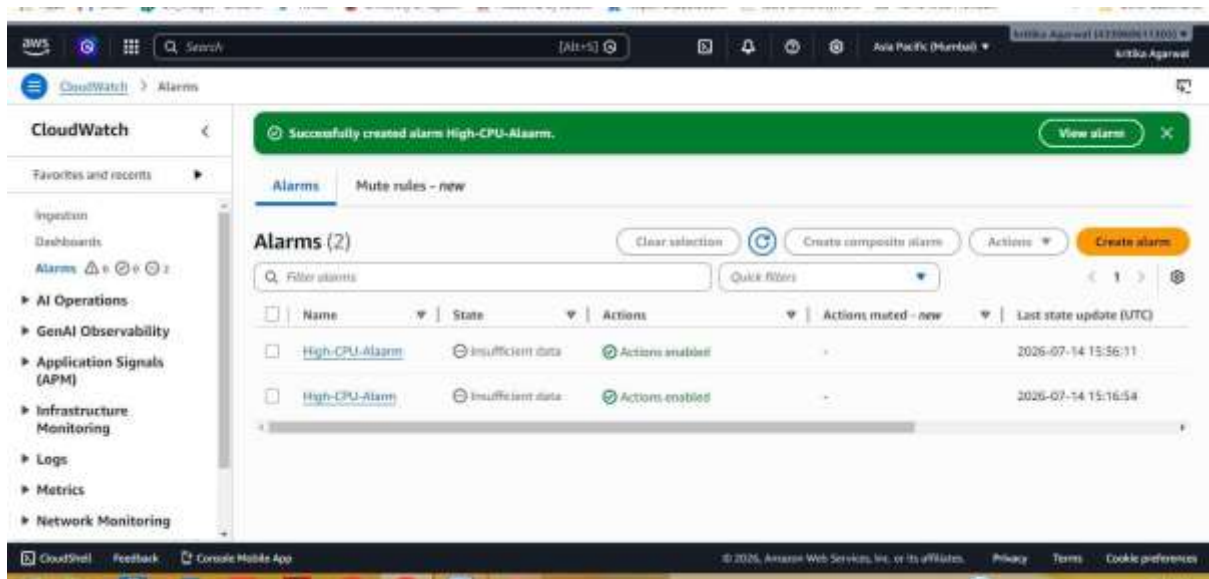


FIG: CLOUD WATCH

### Step 10: Test the Application:

- Open the Load Balancer DNS Name.
- Verify that the website loads successfully.
- Generate CPU load.
- Verify that Auto Scaling launches a new EC2 instance.
- Confirm that CloudWatch triggers an alarm.
- Verify that the SNS notification email is received.



FIG: SUCCESSFULLY CREATE

## 5. CONCLUSION:

### SUMMARY-

- Secure AWS environment is configured.
- VPC and Public Subnets are created.
- EC2 instance is running successful.
- Auto Scaling launches additional instances automatically.
- CloudWatch monitors system performance.

### LEARNING-

- How to configure IAM users securely.
- How to create and configure Amazon VPC.
- How to launch and manage Amazon EC2 instances.
- How to attach and manage Amazon EBS volumes.

# THANK YOU

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